



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
MONDAY TEST

CLASS: X
SUBJECT: PHYSICS

FULL MARKS: 40
DATE: 19.05.2023

General Instructions:

- This paper consists of three printed pages.
- All questions are compulsory.
- Marks will be deducted for spelling errors.

SECTION A

Question 1

Choose the correct answers to the questions from the given options. [1 × 7 = 7]

(i) Two bodies of masses m_1 and m_2 have equal linear momentum. The ratio of their kinetic energies K_1 and K_2 will be

- a) $m_2:m_1$ b) $m_1:m_2$ c) 1:1 d) $m_2^2: m_1^2$

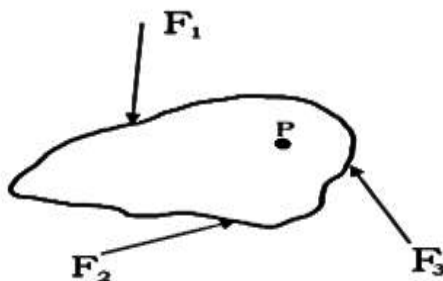
(ii) Choose the correct option wrt the direction of centripetal force.

- a) Towards the centre of the circular path.
b) Away from the centre of the circular path.
c) Normal to the radius at a point where the body is present on the circular path.
d) Direction of this force keeps on changing alternately towards and away from the centre.

(iii) eV/s is the unit of

- a) force b) energy c) power d) mass

(iv) In the figure shown below, three forces F_1 , F_2 and F_3 are seen acting on a body pivoted at P. If each force acts separately and moment of force is same in each case, then



- a) Force F_1 has the maximum value.
b) Force F_2 has the maximum value.
c) Force F_3 has the maximum value.
d) All forces have same value.

- (v) A body is acted upon by two equal and opposite forces along same lines of action of force. The body will have:
- only rotatory motion
 - only translatory motion
 - both (a) and (b)
 - neither (a) nor (b)
- (vi) The energy change during photosynthesis in plants is
- heat to chemical.
 - light to chemical.
 - chemical to light.
 - chemical to heat.
- (vii) The incorrect statement is:
- A body can be balanced at its centre of gravity
 - The total weight of a body acts at its centre of gravity
 - The algebraic sum of moments of the weights of all the particles constituting a body is zero about its centre of gravity.
 - The centre of gravity of a body depends on how it is placed.

Question 2

- (i) a) What is meant by the term 'moment of force'?
 b) If moment of force is assigned a negative sign what will be the turning tendency of the body?
 c) How can moment of force be made zero despite of application of a non-zero finite force? [3]
- (ii) Mention the conditions for work done to be
- maximum
 - negative
- [2]
- (iii) Why are passengers travelling in a double decker bus allowed to stand in a lower deck, but not in the upper deck? [2]
- (iv) A uniform metre rule of weight 10 gf is pivoted at its 0 mark.
- What is the moment of force that depresses the rule?
 - How can it be made horizontal by applying a least force? [2]
- (v) The power of a motor is 40 kW. At what speed can it raise a load of mass 2000 kg? [g = 10 m/s²] [2]
- (vi) A force is applied on a pivoted body. Mention the ways of changing the direction of rotation of the body. [2]

SECTION B

(Attempt all questions.)

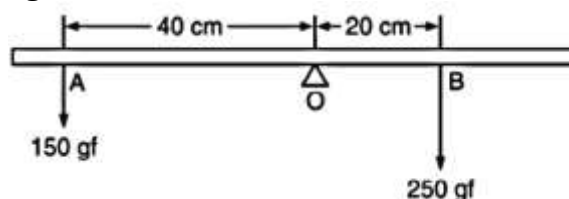
Question 3

[3 + 3 + 4 = 10]

- A force of 100 N acting on a body of mass 20 kg accelerates it uniformly for 10s. If the initial velocity of the body is 5 m/s. Find the work done by the force in 10s, using the concept of work energy theorem.
- Show graphically the variation of different forms of mechanical energy with height, in motion under gravity.
- What is a couple force?
 - Write the expression for moment of couple explaining the term mentioned in the expression.
 - Why does the temperature of water at the bottom of a waterfall differ from the temperature at the top?

Question 4**[3 + 3 + 4 = 10]**

- i) The diagram shows a uniform metre rule weighing 100 gf, placed horizontally on a fulcrum at its centre O. Two weights 150 gf and 250 gf hang from the metre rule as shown in the figure.



- a) Will the rule be in equilibrium?
b) Justify your answer showing relevant calculation.
- ii) A body of mass 0.5 kg falls from a height of 5 m. How much mechanical energy does it possess at any instant? What will be the speed of the object just before it touches the ground? (take $g = 10 \text{ m/s}^2$)
- iii) A stone of mass 'm' is rotated in a circular path with a uniform speed by tying a strong string with the help of your hand. Answer the following questions:
a) Is the stone moving with a uniform or variable speed?
b) Is the stone moving with a uniform acceleration? In which direction does the acceleration act?
c) What is the work done at any instant during the motion mentioned above?